

Effect of stearic acid on rheological properties and printability of ethylene vinyl acetate based feedstocks for fused filament fabrication of alumina

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ABSTRACT

Alumina ceramic feedstocks with ethylene vinyl acetate and stearic acid as an organic binder were prepared and shaped by a commercial 3D filament printer. Rheological properties and the ability of ceramic feedstocks to be processed into filaments and shaped by the fused deposition modeling/filament (FDM/FFF) technique were investigated.

The addition of stearic acid affects the viscosity as a function of shear rate in a complex way. Analysis with rheological models shows that while using a small amount of stearic acid, a viscosity plateau at low shear rate (cross model) can be observed. At high stearic acid content, a yield point (Herschel-Bulkley model) occurs, as the stearic acid content surpasses the amount needed to cover the powder surface. The stearic acid also influences the properties of the solidified filament, making it more brittle and less flexible.

Thin wall structures were printed, debinded and sintered to demonstrate the shape stability and fusion between the layers. Ring-on-ring bending tests of sintered discs show that the printing defects are the primary concerns that determine the strength of sintered samples.

1. Introduction

Fused filament fabrication (FFF) is a well-established additive manufacturing technique. Thermoplastic material is melted and selectively extruded through a nozzle and deposited layer by layer based on the thermoplastic extrusion technique. The ceramic material is fed to the printer in the form of a thermoplastic filament, which is pushed into the hot zone via a roller drive. The advantages include inexpensive equipment, a large variety of materials, easy to use and the possibility of making multi-material and large parts [1]. One of the crucial points, determining the success of the process is the filament properties. The filament must possess certain mechanical properties in order to be suitable for the FFF printer. It must be strong and hard enough to avoid shearing due to pinching from the drive wheel [1] and also stiff enough to avoid buckling between drive wheels and the hot zone of the extrusion die. It should also be noticed that the melting behavior of the material, as well as processing parameters, have a significant influence on successful printing.

The FFF process of ceramics is more complex than the filament printing of a pure thermoplastics or polymer with a low filler content (typically < 20 vol% of inorganic filler). In contrast to polymers with filler materials, the powder loading for ceramic feedstock is relatively high, typically between 45 and 60 vol%. The high filling level is needed

to ensure proper sintering of the ceramic in the final processing step. The thermoplastic component and other polymeric processing additives like softeners and surfactants, which are essential for the shaping process, have to be completely removed before the sintering step. Therefore, a so-called debinding step is necessary.

As already mentioned, feedstocks with high solid loading content are more challenging for processing since they are more brittle and difficult to feed through the nozzle [2]. There are several literature that performs real time monitoring of print quality using optical imaging of each layer [3], acoustic emission technique [4], vibration sensors [5] etc. The 3D printing of ceramic filament depends on the rheological behavior of the feedstocks, the mechanical properties of the filament, the geometry of the filament, the design of the machine, the processing parameters and the printing head [1]. If the resistance to flow through the heated nozzle is too high, the filament will either buckle or the driving rollers will just slip or grind the surface of the filament without pushing the filament further. The buckling is a common failure mode of filaments with ceramic fillers [6]. Insufficient filament stiffness or too high viscosity results in the buckling of filament [2,7], or blocking inside the nozzle [8]. A relation between critical buckling stress and different filament properties is given in literature [7]. A decrease in filament diameter decreases the critical buckling stress and causes buckling. Similarly, an increase in filament diameter restricts the entry

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of filament into the entrance. Hence, a tight tolerance of filament diameter is crucial for the unhindered extrusion. Feeding rate and the buckling during the FFF process limits the composition of the feedstock recipe used for the filament production [6]. However, the filament should be flexible enough to be spooled, so that it can be easily stored in a compact place and fed in a continuous manner [1]. In addition, the filament must be bent between the spool and the printing head, which is another reason why certain flexibility of the filaments is required. Another important parameter that determines the efficiency of the filament to be extruded without buckling is the ratio between compressive modulus, E to the apparent viscosity, η_a . It was found that feedstocks which exhibited a value above the critical ratio, (E/η_a) which falls in the range of $3 \times 10^5 - 5 \times 10^5 \text{ s}^{-1}$ could be printed without buckling [9].

Several formulations for the ceramic filaments were reported by researchers for different materials like tricalcium phosphaste TCP [10], [11,12], mullite [13], alumina [10,14,8], zirconia [16,30,14], silicon nitride [17,7], and PZT [18,19]. Nowadays, sintered ceramic parts made by the FFF does not achieve the strength obtained by other thermoplastic shaping methods which could be considered as a limiting factor of this technique.

In this work, the effect of stearic acid content on the rheological properties of the ceramic feedstocks was analyzed. Well-established rheological models were used to describe the flow behavior of feedstocks. In order to find the optimized stearic acid content, the vase structure was printed with a commercial filament printer. Finally, disc structures were printed and sintered to measure the biaxial bending strength, and the microstructure was also examined.

2. Experimental

For this study, Alumina CT3000 (Almatis GmbH, Germany) and ethylene vinyl acetate EVA 420 (DuPont, USA) were used for the formulation of the ceramic feedstock. The CT3000 is a 99.8 % pure Al_2O_3 powder with a median particle size of 0.5 μm and BET specific surface area of 7.8 m^2/g . The EVA copolymer has a relatively high melt flow index (150 g/10 min at 190 °C). As a surfactant, stearic acid (Sigma Aldrich, Switzerland) was added in different concentrations (0–17 wt. %). To investigate the rheological behavior and printability of the ceramic filaments, ceramic powder content was kept constant to 50 vol. %. Thermoplastic feedstocks were prepared by mixing all components in a high shear mixer (Rheomix 600, HAAKE PolyLab OS, Thermo Electron Corporation, Germany). Before mixing, alumina powder CT3000 were dried overnight at 120 °C. The mixing was done using roller rotors at 120 °C and 10 rpm for 60 min.

Two different rheological tests were made at 150 °C using a rotational rheometer (MCR 302, Anton Paar, Austria) with a plate on plate configuration with a 0.5 mm gap. To investigate the viscosity, the shear stress was measured by a given shear rate program. To investigate the yield point, the shear rate was investigated by controlling the shear stress. In the second test, the yield point can be determined by measuring the shear stress value where the material starts to flow (shear rate starts to increase rapidly). For the FDM/FFF process, filaments were made by extrusion of the feedstocks at 90 °C through a die orifice with a diameter of 1.75 mm, using a piston extruder (RH7 Flowmaster, Malvern Instruments Ltd, UK). A commercial FDM/FFF printer Craftbot 2 (Craft Unique, Hungary) was used for the fabrication of the vase and disc structures. The nozzle diameter was 0.6 mm, the printing speed was set to 10 mm/s and the nozzle temperature was varying between 130 °C – 170 °C. Feedstocks with 0 and 17 wt.% stearic acid were investigated by simultaneous thermal analysis (TGA/DSC) with a heating rate of 5 K/min in air atmosphere (STA 449 F3 Jupiter, Netzsch, Germany), and by thermo-mechanical analysis (TMA) in air (TMA 402 F3 Hyperion, Netzsch, Germany).

Based on the STA analysis and previous work [7,14], thermal debinding and pre-sintering programs were carried out in static air using a

Table 1

List of all feedstocks. All feedstocks contained 50 vol % of Al_2O_3 powder.

Feedstock	wt% binder [%]	Binder composition		
		Elvax 420 [%]	stearic acid [%]	m/A [mg/m^2]
1	19.0	100.0	0.0	0.0
2	19.0	99.0	1.0	0.3
3	18.9	97.9	2.1	0.6
4	18.9	96.5	3.5	0.9
5	18.9	95.8	4.2	1.3
6	18.9	94.7	5.3	1.6
7	18.7	92.0	8.0	2.4
8	18.7	83.0	17.0	5.0

furnace (LT 40/12, Nabertherm GmbH). For the debinding step, dwell time at 230 °C, 375 °C and 500 °C was used. After debinding the samples are too fragile to handle and therefore a pre-sintering step at 1000 °C was added to the debinding program. The final sintering step was done in an electrically heated furnace (LHT 04/17, Nabertherm, Germany) at 1600 °C, with a heating rate of 5 K/min and dwell time of 1 h.

The strength of sintered 3D printed discs were evaluated using a standard ring-on-ring bending test on cylindrical samples [20]. Based on the thickness of the sample, a support ring with a diameter of 18 mm and a load ring with a diameter of 7 mm were selected. For the mechanical analysis, a universal tensile testing machine Zwick Z005 (ZwickRoell GmbH & Co. KG, Germany) was used. The microstructure of the fracture surface was investigated by scanning electron microscopy (SEM, VEGA3, Tescan, Czech Republic).

3. Results

3.1. Materials and feedstocks formulations

Eight different feedstocks with different amounts of stearic acid were investigated. The compositions are listed in Table 1.

Table 1: List of 8 feedstocks with different stearic acid content. All feedstocks contained 50 vol % of Al_2O_3 powder. Because of the different density of Elvax 420 and stearic acid, the weight content of the binder differs.

The (m_{SA}/A) ratio can be calculated from feedstock compositions according to Eq. 1.

$$\frac{m_{SA}}{A} = \frac{w_B - w_{SA,B}}{(1 - w_{SA,B}) BET} \quad (1)$$

where w_B is the percentage of all organic binders in the feedstock (second column in Table 1), $w_{SA,B}$ is the percentage of stearic acid in the binder (third column in Table 1) and BET is the specific surface area of the CT3000 alumina powder.

3.2. Rheological properties

Viscosity as a function of shear rate shows shear-thinning (pseudoplastic) behavior (Fig. 1). When stearic acid content is increased from 0 % to 3.5 % the viscosity is uniformly decreased at all shear rates. However, between 3.5 and 4.2 %, a drastic change in the flow behavior appears. In order to better understand these changes the flow curves were fitted by common rheological models like Power-law (eq. 2), Herschel-Bulkley (eq. 3), Cross (eq. 4) and Carreau/Gahleitner (eq. 5).

Power-law is a very simple, but often effectively used two-parameters rheological model. Herschel-Bulkley model is a widely used three-parameter rheological model for describing the viscosity as a function of shear rate of fluids with yield point. It is commonly used for suspensions with a high concentration of particles, which typically show shear thinning and a yield point behavior [21]. Cross and Carreau/Gahleitner models are typically used for polymers with low to moderate concentration of particles [21].

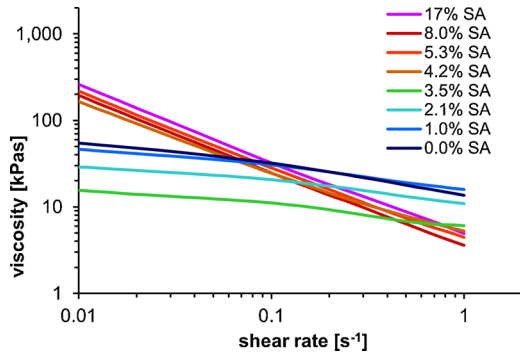


Fig. 1. Viscosity as a function of shear rate for feedstocks with different stearic acid amount system measured at 150 °C.

$$\eta = k \dot{\gamma}^{n-1} \quad (2)$$

$$\eta = \frac{\tau_0}{\dot{\gamma}} + k \dot{\gamma}^{(n-1)} \quad (3)$$

$$\eta = \frac{\eta_0}{1 + (c \dot{\gamma})^p} \quad (4)$$

$$\eta = \eta_{inf} + \frac{(\eta_0 - \eta_{inf})}{(1 + (c \dot{\gamma})^p)^q} \quad (5)$$

where η is the viscosity, τ_0 , k , n , c , p , η_0 , η_{inf} and q are model parameters.

The goodness of fit was evaluated by the coefficient of determination (R^2) and the comparison for different models is shown in the Fig. 3. The well-known R^2 was calculated according to Eq. 6 using the logarithmic values of viscosities.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - a_i)^2}{\sum_{i=1}^N (y_i - a_{mean})^2} \quad (6)$$

where N is the number of measuring points, y_i are the fitted values according to the certain rheological model, a_i are the measured values and a_{mean} is the mean of measured values. Best fitting parameters for maximizing the R^2 were found with the Microsoft Excel Solver software using GRG Nonlinear method.

Depending on the stearic acid content, two distinctive regions were observed in which models have a significantly different fitting. Between 0 % and 3.5 % Cross and Carreau/Gahleitner show better fitting compared to Herschel-Bulkley and Power-law (Fig. 2a). The last two show identical values because the best fitting (between 0 % and 3.5 % stearic acid) was found using a yield point of 0. It is worthwhile to mention that by removing the yield point parameter in the Eq. 3, obviously Eqs.

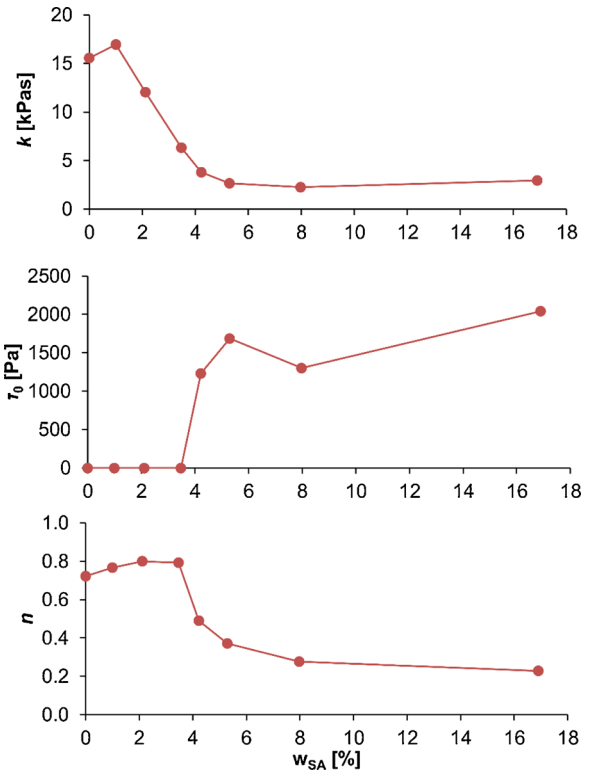


Fig. 3. Herschel-Bulkley parameters obtained from best fitting to the viscosity curves for feedstocks with different amounts of stearic acid in the thermoplastic binder composition.

2 and 3 become the same. Carreau/Gahleitner model (having 5 parameters) fits only slightly better than the Cross model (having 3 parameters). Cross model has also the same number of parameters as Herschel-Bulkley, which helps for the direct comparison. Therefore, the Cross model that was preferred over the Carreau/Gahleitner model. Cross model can be actually considered as a simplified Carreau/Gahleitner model (when $\eta_{inf} = 0$ and $n = 1$).

For feedstocks with equal or more than 4.2 % of stearic acid, a relatively good fitting according to the R^2 is obtained for all models. This is hardly surprising since zero viscosity for Cross and Carreau/Gahleitner model is extrapolated using the last measured point at low shear rates. It should be noted that the viscosity measurements at higher shear rates are not possible due to slipping effect.

For comparison, both of the 3 parameter models, e.g. Herschel-

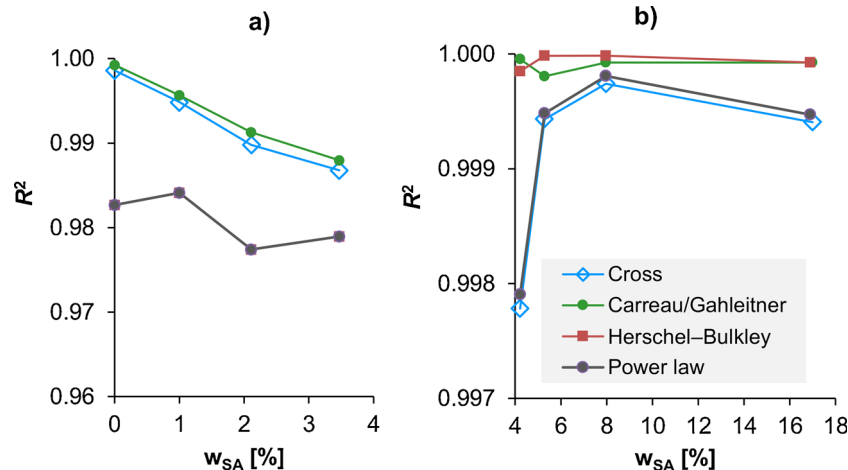


Fig. 2. The goodness of fit for measured viscosity as determined by the R^2 .

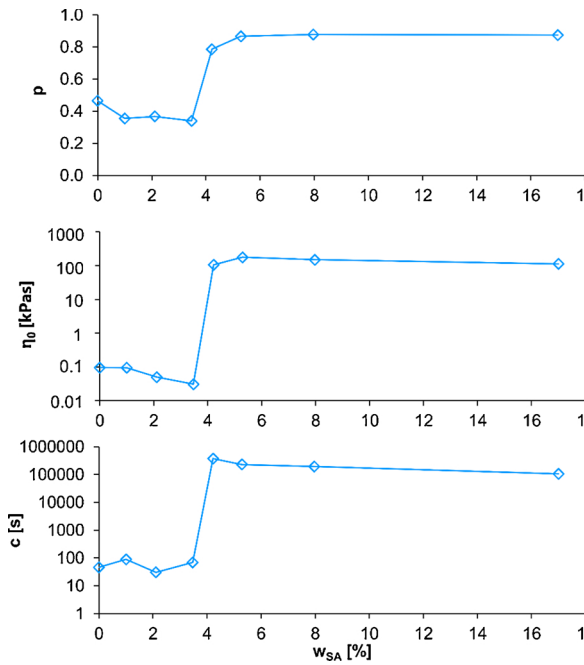


Fig. 4. Cross model parameters obtained from best fitting to the viscosity curves for feedstocks with different amounts of stearic acid the thermoplastic binder composition.

Bulkley and Cross model were further investigated. The 3 different parameters of both models are shown in Figs. 3 and 4. In Fig. 3 the Herschel-Bulkley k parameter, which is associated with the general scale of viscosity, is decreasing with the addition of stearic acid and reaches a plateau of low values at around 5 % stearic acid. The yield point parameter (τ_0) is 0 Pa for feedstocks with stearic acid content between 0 % and 3.5 % and significantly increases when stearic acid content surpasses 4.2 % with values calculated between 1200 Pa and 2050 Pa. The n parameter drastically decreases when stearic acid content increased to 4.2 % and reaches a plateau of low values at around 8 % of stearic acid. Low n value (< 1) indicates a shear-thinning behavior; the lower the n , the more pronounced is the shear thinning effect.

In the Cross model (Fig. 4), parameter p indicates a shear-thinning behavior. A value of zero indicates a Newtonian behavior and the higher the p , the more pronounced is the shear thinning behavior. Therefore, the shear-thinning behavior becomes more pronounced with increased stearic acid content, which is in good agreement with the previously discussed results achieved by using the Herschel-Bulkley model. Parameter η_0 , which is associated with viscosity plateau at low shear rates, increases to extremely high values when more than 4.2 % stearic acid is used in the feedstock. The parameter c is associated with the onset of shear-thinning behavior. The reciprocal value ($1/c$) represents a critical shear rate of the onset for shear thinning. Very high values of c mean that a transition to shear thinning behavior starts at extremely low shear rates. This confirms that above 4.2 % of stearic acid shear-thinning already starts at very low shear rates.

Fig. 5 shows the result of the second rheological investigation where shear rate was measured by controlling the shear stress. The results confirmed that for the feedstocks with a stearic acid content of 3.5 % or less, no yield point could be detected. In the case of feedstocks with stearic acid amounts higher than 4.2 %, the feedstocks start to flow after a certain threshold shear stress (the yield point). The transition is not entirely sudden, however, shear stress values at which the feedstocks begin to flow are in a similar range as those determined by fitting viscosity curves using Herschel-Bulkley model (Fig. 3). Better accordance could be expected if rheological measurements at higher shear rates could be achieved without slip.

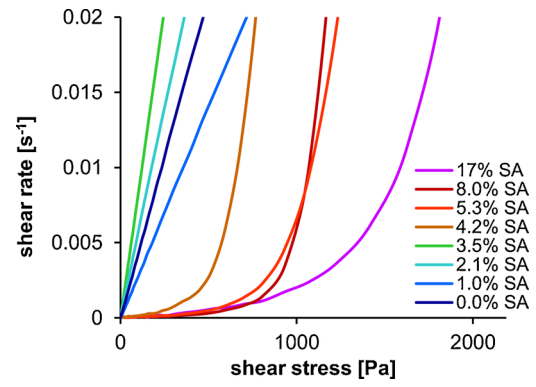


Fig. 5. Shear rate as a function of shear stress, measured by a plate on plate system measured at 150 °C.

The yield point is important for shape retention during the de-binding process. Objects made from feedstocks with zero yield point tend to collapse and deform during the thermal debinding process [11]. Therefore, feedstocks with a certain yield point are favorable for the filament printing of ceramic structures.

The 4.2 % of stearic acid corresponds to 1.3 mg/m^2 of (m_{SA}/A) ratio (Table 1). It has been reported that one molecule of stearic acid theoretically covers 0.2 nm^2 of the surface [22–25] and the amount of stearic acid for fully saturated monolayer coverage can be calculated according to the Eq. 7 [22].

$$FSM = \frac{M_{SA}}{N_A S_{SA}} \quad (7)$$

where $FSM [\text{mg/m}^2]$ is the amount of stearic acid at a fully saturated monolayer of stearic acid on the powder surface, M_{SA} is the molecular weight of stearic acid, N_A is the Avogadro constant and S_{SA} is the area coverage of one stearic acid molecule (0.2 nm^2). According to Eq. 7, a fully saturated monolayer of stearic acid (FSM) would be expected at 2.27 mg/m^2 of stearic acid per surface of the powder. This is close to 8 % of stearic acid in the binder – see Table 1. However, a significant change in rheological behavior could be observed at 4.2 %, which corresponds to 1.3 mg/m^2 of (m_{SA}/A) ratio. This is less than the theoretical value for a monolayer. This discrepancy can be explained if the surface, covered by the stearic acid, is actually smaller than that measured by BET. Stearic acid covers the actual particles and aggregates. Useful quantity, which connects the agglomerated particle surface with that measured by BET is the agglomeration factor. The agglomeration factor (F), which describes the ratio between the measured average particle size d_{50} and the d_{BET} can be calculated from the specific surface area (BET) and density according to Eqs. 8 and 9 [22].

$$F = \frac{d_{50}}{d_{BET}} \quad (8)$$

$$d_{BET} = \frac{6}{BET \rho} \quad (9)$$

For the CT3000, the agglomeration factor is 2.6. It is expected that during the compounding the agglomerates cannot be destroyed and therefore a much lower stearic acid content is needed to cover the total surface of the alumina powder. If the average particle size d_{50} of the powder was used to calculate specific surface area, based on Eq. 9, a much lower BET (3 g/m^2) will be observed. Using average particle size d_{50} and the density of the CT3000, stearic acid content of 3.2 % is needed to build up a monolayer around the agglomerated alumina powder (Eq. 7). This theoretically calculated value is very close to the stearic acid content 4.2 %, where a drastic change in the flow behavior of the ceramic feedstock can be observed. Therefore, by using apparent BET of the agglomerated powder, the amount of stearic acid to achieve the apparent saturation can be estimated. Below the apparent

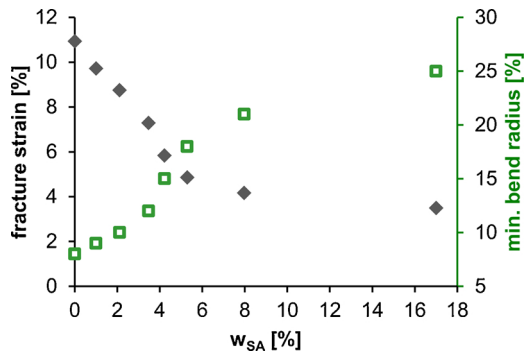


Fig. 6. Fracture strain of the filaments, calculated by Eq. 10 (grey diamond markers) and min. bending radius at which the filament broke (green hollow squares).

saturation, the stearic acid binds to the surface of the ceramic powder and acts as a surfactant. Above the apparent saturation, some free stearic acid exists that remains unattached to the particles. An excessive amount of stearic acid can be integrated into the polymer matrix and reduces the overall viscosity of the feedstock [26–28].

3.3. Filament flexibility

A “flexibility test” was proposed to predict the handling of filaments in FFF printers. This test measures the maximal curvature to which a filament can be bent, before it fractures. A bend radius is recorded and strain at the outer edge of filament is calculated according to Eq. 10.

$$\varepsilon = \frac{d}{2R} \quad (10)$$

where ε is fracture strain, d is the filament diameter and R is the minimum radius of curvature. The results of this flexibility test are shown in Fig. 6. Filaments become less flexible with the increased amount of stearic acid (= higher bending radius at breaking point) and therefore the fracture strain where the filaments break decreases. Between 4.2 % and 5.3 % of SA, the bending radius and fracture strain started to reach a plateau (constant value). At this amount of stearic acid, the flow behavior of the feedstock also changes and a yield point appears. At 17 % of stearic acid, the filament can already break easily during the handling. Higher amounts of stearic acid would make the material even less flexible and thus impractical for handling.

As already mentioned above, it is assumed that stearic acid preferentially binds to the surface of ceramic particles. At small addition of stearic acid (< 4.2 %), it spreads over a large surface area of particles. Flexibility as a function of stearic acid concentration rapidly decreases at low amounts of stearic acid (Fig. 6). After reaching apparent saturation and all particles are coated (at around 4.2 % stearic acid), free stearic acid starts to accumulate in the EVA matrix, and form regions or

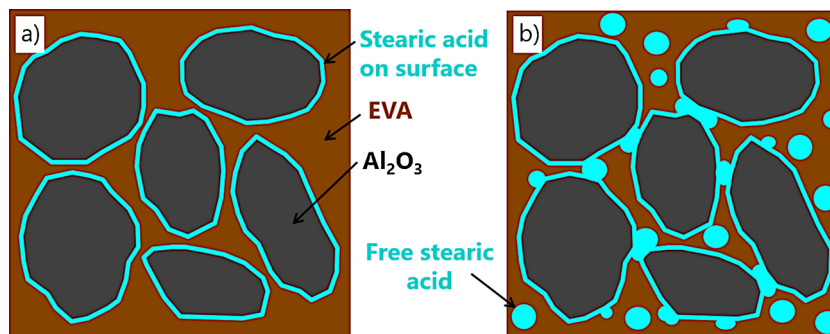


Fig. 7. Sketch to explain the distribution of stearic acid in the feedstocks a) below the apparent saturation on the powder surface and b) above the apparent saturation when free stearic acid forms regions or networks in the EVA matrix.

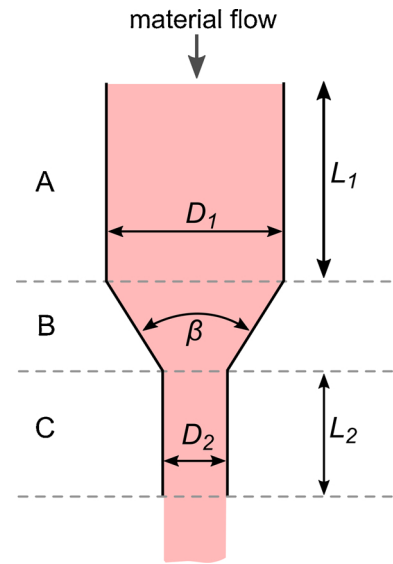


Fig. 8. Sketch of extruder divided into three zones for modelling.

Table 2

Parameters of the power-law model (Eq. 2) used for the calculations of force need to push the filament through the nozzle (Eqs. 11–15)

stearic acid [%]	k [Pas]	n
0.0	15,560	0.72
1.0	16,942	0.77
2.1	12,026	0.80
3.5	6319	0.79
4.2	4337	0.21
5.3	3998	0.13
8.0	3387	0.12
17	4593	0.13

network inside the EVA matrix. The additional stearic acid causes the yield point rheological behavior. A sketch to describe this phenomenon is shown in Fig. 7. Similar effects were also observed for tricalcium phosphate feedstocks [11].

3.4. Shaping

One of the most important features of the filament is printability, i.e. the ability of the filament to be processed by the 3D printer and produce desired shapes without interruption. Design and printer settings play a crucial role in successful printing, so the evaluation of filament's printability is not trivial. Generally, slower printing speeds result in a more stable process with better quality. A speed of 10 mm/s was used for the printing, which is a reasonable compromise between

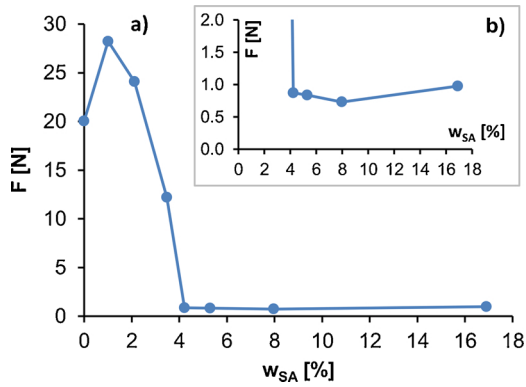


Fig. 9. Force needs to push the filament through the nozzle at speed 10 mm/s and temperature 150 °C, according to the Eqs. 11 – 15. In a) full scale is shown and in b) low forces are shown for better clarity.

the quality and the printing time. In this study, the effect of the extruder head temperature on the printing quality was also investigated. Hence, all the filaments were printed with a nozzle temperature between 130 °C and 170 °C.

Based on the rheological calculation shown in Figs. 3 and 4, the force needed to extrude the filaments with different stearic acid content was calculated. A model for calculating the pressure drop inside the printing head of a filament printer was described by Ramanath et al.

[29] and by Turner et al. [6]. Both contributions assume an incompressible liquid, no slip boundary conditions, and laminar flow. To describe the behavior of the molten filament inside the printer head power-law model (Eq. 2) was used. The pressure drop is calculated for each zone (A, B and C) in the printer head (see Fig. 8) using the Eqs. 11–15. The total pressure drop (ΔP) is the sum of pressure drops in each zone (Eq. 14). The force needed to push the filament is the product of the total pressure and the cross-section area of the filament (Eq. 15).

$$\Delta P_A = 2 L_1 \left(\frac{v}{k^n} \right)^{-n} \left(\frac{-\frac{1}{n} + 3}{(D_1/2)^{-\frac{1}{n}+1}} \right)^{-n} \quad (11)$$

$$\Delta P_B = \left(\frac{-2}{n 3 \tan(\beta/2)} \right) \left(\frac{1}{D_2^{-3n}} - \frac{1}{D_1^{-3n}} \right) \left(\left(\frac{D_1}{2} \right)^2 \left(-\frac{1}{n} + 3 \right) 2^{-\frac{1}{n}+3} \right)^{-n} \quad (12)$$

$$\Delta P_C = 2 L_3 \left(\frac{v}{k^n} \right)^{-n} \left(\frac{\left(-\frac{1}{n} + 3 \right) (D_1/2)^2}{(D_2/2)^{-\frac{1}{n}+3}} \right)^{-n} \quad (13)$$

$$\Delta P_{tot} = \Delta P_A + \Delta P_B + \Delta P_C \quad (14)$$

$$F = \Delta P_{tot} \left(\frac{\pi D_1^2}{4} \right) \quad (15)$$

where v is the entry speed of the filament, k and n are parameters calculated by the power-law model (Eq. 2), L_1 , L_2 , D_1 , D_2 and β are

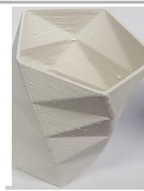
SA [wt%]	130°C	140°C	150°C	160°C	170°C
0.0	—	—	—	—	—
1.0	—	—	—	—	—
2.1	—	—	—	—	—
3.5	—	—	—	—	—
4.2	—	—	—		
5.3	—	—			
8.0	—	—			
17					

Fig. 10. A summary of the printability test for feedstocks with different stearic acid content (SA). A constant printing speed of 10 mm/s and a nozzle diameter of 0.6 mm were used. Dash line (—) indicates that filament was not printable at those conditions (filament was blocked in the printing head).

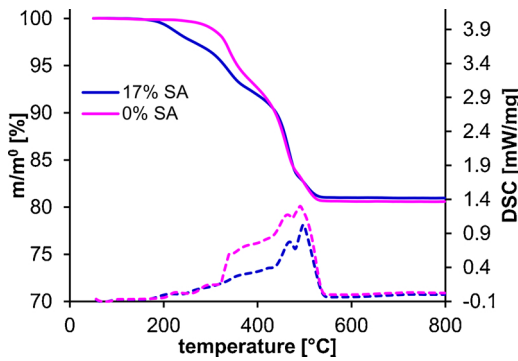


Fig. 11. Thermal analysis of two different feedstocks (with 0 % and 17 % stearic acid) made with a heating rate of 5 K/min in air. The full lines present TG and the dotted line presents DSC.

geometrical parameters of the extruder (Fig. 8). For the calculation, $v = 10$ mm/s, $L_1 = 14$ mm, $D_1 = 1.75$ mm, $L_2 = 2$ mm, $D_2 = 0.6$ mm and $\beta = 60^\circ$ were used. The parameters k and n , used in the calculation, were determined from fitting the rheological curves (Fig. 3). The parameters are listed in Table 2.

In Fig. 9 the results of the calculated force needed to push the filament through the nozzle are presented.

The forces calculated by the Ramanath's model give very interesting results. A huge decrease in the required force can be observed when 4.2 % or more of stearic acid is added. Compositions with stearic acid content between 4.2 % and 17 % show more or less similar required force. To verify the results from the modeling, printability tests were performed at temperatures between 130 and 170 °C (Fig. 10). The results correspond well with the Ramanath's model. It was not possible to print vases with a stearic acid content below 4.2 %. The filaments blocked and could not be pushed through the nozzle –too large force to push the filament through the nozzle was required (Fig. 9). However, the model predicts no significant differences between feedstock with stearic content between 4.2 %–17 %. Nevertheless, in practice, differences in the quality of the printed products were observed when stearic acid content was increased. As already mentioned, a stearic acid content above the apparent saturation will result in free SA inside the EVA and slip in the printing nozzle can be expected. Therefore, the model of Ramanath will not be valid any longer. Feedstock with 17 % of stearic acid was considered the best to print because it could be processed in the largest temperature range. It can be expected that by increasing the temperature inside the printing nozzle, the force to push the filament through the nozzle will decrease. Therefore, it can be assumed that the force to print filaments with higher stearic acid content will decrease.

3.5. Debinding and sintering

Thermal debinding presents a major challenge for the 3D printed part. Shape collapse, cracks, bloating and delamination can occur due to the binder decomposition. Stearic acid content can significantly affect the thermal debinding process [18]. Therefore, thermal analysis of feedstock without stearic acid (0 % SA) and the highest amount of stearic acid (17 % SA) were investigated and are shown in Fig. 11. Stearic acid begins to decompose at 200 °C with minor exothermic effect, which suggests oxidation is not very dominant and evaporation is an important mechanism of decomposed organic gas transport. This is in good agreement with the kinetic results reported by Salehi et al. [22]. At around 300 °C EVA starts to decompose with a major exothermic reaction along with a mass loss peak at 450 °C. The main difference in the TG/DSC curve of the feedstock without stearic acid and a high amount of stearic acid can only be detected at a temperature between 200 and 300 °C. Therefore, dwell time at 230 °C was selected to avoid cracking during debinding of printed structures with higher stearic acid content. Due to the high mass loss rate at 375 °C, a second dwell time was selected. It is worthwhile to mention that a high content of stearic acid is beneficial for the thermal debinding process since stearic acid can be removed earlier than the EVA and thus a gradual binder removal can be achieved.

Vase structures at different processing stages are shown in Fig. 12. No shape collapse or deformation was observed during the thermal debinding step. A total shrinkage of 23 % in the height, 18 % in the diameter and 12 % the wall thickness was calculated by the analysis of the green and sintered vases.

It is known that binder systems with EVA tend to form a dense dark skin during the thermal debinding in air atmosphere [11]. The formation of a dense skin during oxidative decomposition of EVA in the shaped body limits the thickness of the samples that is possible to debind in a practical process. Because of this limitation, dense ceramic discs with 2 layers (0.50 mm in the sintered state) and 4 layers (1.05 mm in the sintered state) and a diameter of 20.0 mm were printed, debinded, and sintered to investigate the biaxial strength using ring-on-ring test configuration. After sintering, the Archimedes density for 2 and 4 layers were 3.95 g/cm³ and 3.90 g/cm³, respectively. It is a challenge to make full dense samples with FFF technique [8]. The shrinkage of the discs was 23 % in height and 17 % in diameter for both sizes. The mechanical ring on ring test is a biaxial test, where two directions are stressed simultaneously (unlike the uniaxial 4-point bending test). This is quite practical for 3D printed parts since samples are characteristically anisotropic due to the printing directions. The tensile stressed side during the test was the first printed layer (touching the substrate during printing). The mechanical tests show that strength

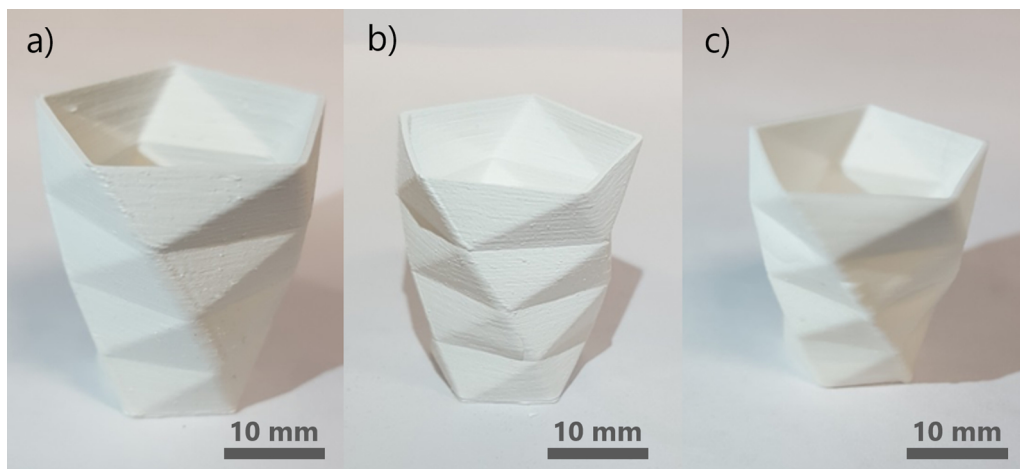


Fig. 12. Vase structure during different processing stages: a) After printing, b) after debinding and pre-sintering at 1000°C and c) after sintering at 1600°C.

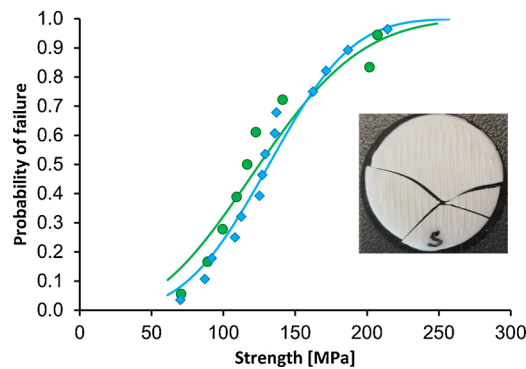


Fig. 13. Ring on ring strengths of sintered ceramic discs with 2 or 4 layers thickness (points). Based on the results, characteristic strength and Weibull modulus were 147 MPa and 3.3 for 2 layered discs and 144 MPa and 2.6 for 4 layered discs, respectively.

values were relatively low for alumina, as seen from Weibull distributions in Fig. 13. No significant differences in strengths were observed between the samples made from 2 or 4 layers.

Microstructure analysis of fracture surfaces were performed on selected discs after the mechanically testing. In Fig. 14a, a 4 layered disc with lowest flexural strength (89 MPa) is shown. Printing defects are clearly visible in the middle of the printed discs. As expected, the fusion between layers was very good and no interface can be seen inside the microstructure (Fig. 14b). The fracture surface of the 4 layered disc with the highest strength (207 MPa) is shown in Fig. 14c. Some small printing defects were detectable, but on the compressive side during the test. To investigate the grain structure of the disc with the highest strength, higher magnification was used (Fig. 14 d). Smaller pores inside the grains and at the grain boundaries can be observed.

4. Conclusions

Thermoplastic Al_2O_3 filaments were prepared using ethylene vinyl

acetate and stearic acid as an organic binder. Stearic acid concentration significantly affects the rheological properties of the thermoplastic feedstocks and printing behavior of thermoplastic filaments.

A substantial change in rheological behavior is observed when the apparent saturation of stearic acid on the surface of powder occurs. In general, a yield point can be detected above 3.5 % SA and shear-thinning effect suddenly becomes stronger, which is evident from the parameters of Herschel-Bulkley model. In fact, the changes in flow behavior are so significant, that different rheological models are needed to describe the flow behavior for lower and higher amounts of stearic acid. For feedstocks with a lower stearic acid content, < 4.2 % Cross model result in a better fitting, while for feedstocks with a higher stearic acid content, the Herschel-Bulkley model is needed. Based on rheological characterization, a theoretical prediction of the force needed to push the filaments through the printing head of the FFF printer was performed. In general, the model described by Ramanath et al. [29] and Turner et al. [6] can be used to determine if the ceramic filament will be printable or not. Above the apparent saturation of stearic acid, the model becomes incorrect because of the slipping effect, which is expected for higher stearic acid contents. The flexibility of ceramic filament is reduced by increasing the stearic acid content. However, printability improved for very high stearic acid contents and vase structures could be easily extruded in a wide temperature range of the heated printing nozzle.

To explain the effect of the stearic acid on the flow behavior of the feedstock and the flexible properties of the filaments, a model has been proposed. Stearic acid buildup a monolayer coating on the surface of the ceramic powder. After reaching an apparent saturation on the surface of the powder, free stearic acid starts to build up regions or network inside the EVA matrix. Thus, an increase of the yield point in the feedstocks with higher stearic acid content can be explained. Similar behavior has been previously reported on filaments based on tricalcium phosphate, EVA and SA [10].

The described binder system enables the printing of thin-walled alumina structures that can be successfully debinded and sintered. However, ring-on-ring bending tests revealed quite low mechanical

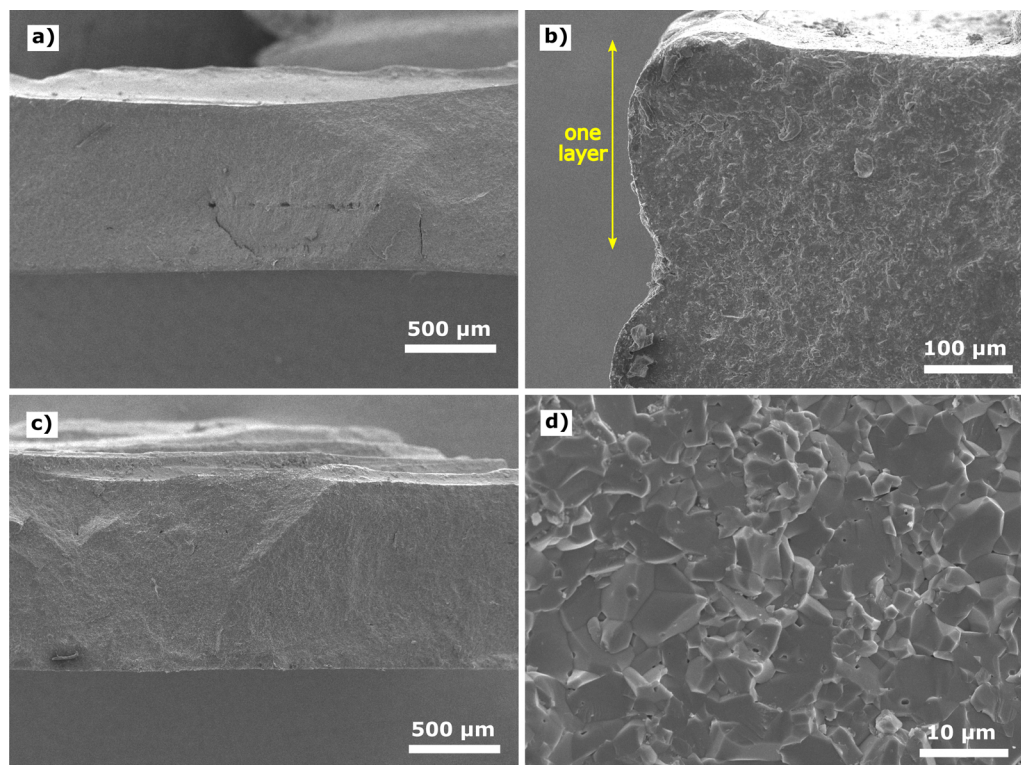


Fig. 14. The microstructure of fractured surfaces after the ring on ring test of two 4 layered discs. a) printing defects of the disc with low strength of 89 MPa, and b) same disc but with higher magnification at the edge to evaluate the fusing of the different layers, c) overview of disc with high strength of 207 MPa and d) high magnification of the same disc to evaluate grain structures and small pores. In the SEM pictures a) and c), the tensile stressed sides are at the bottom.

properties of the sintered ceramic and further investigation has to be done to improve the mechanical values. Mainly printing defects and pores at the grain boundaries could be localized on the fracture surfaces of the mechanical tested printed discs.

CRedit authorship contribution statement

Lovro Gorjan: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Cristina Galusca:** Data curation, Methodology, Validation, Formal analysis. **Marwah Sami:** Methodology, Validation, Formal analysis, Data curation. **Tutu Sebastian:** Data curation, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Frank Clemens:** Conceptualization, Funding acquisition, Methodology, Formal analysis, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

As a first and corresponding author I confirm in the name of all authors, that the authors that have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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